

|                                                                                                                         |                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>(c) Stellar evolution</b>                                                                                            |                                                                                                                                                                                                                                 |
| <b>Students should:</b>                                                                                                 |                                                                                                                                                                                                                                 |
| 8.7                                                                                                                     | understand how stars can be classified according to their colour                                                                                                                                                                |
| 8.8                                                                                                                     | know that a star's colour is related to its surface temperature                                                                                                                                                                 |
| 8.9                                                                                                                     | describe the evolution of stars of similar mass to the Sun through the following stages: <ul style="list-style-type: none"> <li>• nebula</li> <li>• star (main sequence)</li> <li>• red giant</li> <li>• white dwarf</li> </ul> |
| 8.10                                                                                                                    | describe the evolution of stars with a mass larger than the Sun                                                                                                                                                                 |
| <b>8.11P understand how the brightness of a star at a standard distance can be represented using absolute magnitude</b> |                                                                                                                                                                                                                                 |
| <b>8.12P draw the main components of the Hertzsprung–Russell diagram (HR diagram)</b>                                   |                                                                                                                                                                                                                                 |

---